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Key Points:

- Internal variability explains about 40% of observed April Arctic-mean surface air temperature trend over 1980–2022
- The multi-model mean robustly captures monthly externally forced Arctic amplification and warming patterns in March, April, and May
- Current climate models may underestimate internal variability in the pressure-temperature trend configuration in March, April, and May

Supporting Information:

Supporting Information may be found in the online version of this article.

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Why April Stands Out: Monthly Impacts of Internal Variability on Arctic Amplification

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Abstract Observed Arctic amplification (AA), defined as the ratio of Arctic-mean to global-mean surface air temperature (SAT) trends, peaks in April, a feature not captured by climate models. Here, we quantify the impact of internal variability on Arctic-mean SAT trends in March, April, and May from 1980 to 2022 using multiple approaches, including machine learning algorithms—convolutional neural networks (CNNs) and partial least squares (ML-PLS)—as well as dynamical adjustment and wind-nudged experiments. Across all approaches, internal variability emerges as a substantial contributor to Arctic warming in these months, and is particularly prominent in April, when internal variability accounts for $40 \pm 31\%$ of the observed Arctic-mean SAT trend, a result consistent across the CNN and wind-nudged estimates. For each spring month, the linear methods estimate 4%–15% smaller contributions of internal variability than CNN estimates, suggesting that nonlinear approaches may more effectively isolate the internal variability component. Removing internal variability largely reconciles the springtime monthly AA discrepancy between climate models and observations. The strong spatial correlation between the observed-minus-multi-model-mean SAT trend pattern and that derived from wind-nudged experiments further suggests that the simulated forced trend pattern is broadly realistic. Finally, the observed springtime monthly circulation–temperature configurations are rare in coupled climate models across all three spring months, suggesting that models have difficulty reproducing the magnitude of Arctic internal variability and/or the observed patterns are a rare manifestation.

Plain Language Summary Arctic warming outpaces global warming, especially in spring, a pattern climate models fail to reproduce. This study examines how much of the March, April, and May warming trends during 1980–2022 are due to natural variability. Using several independent methods, including machine learning (neural networks and a linear technique) and physics-based approaches (dynamical adjustment and wind-nudged experiments), we find natural variability plays a substantial, month-dependent role in springtime Arctic warming. The effect is strongest in April, where natural variability explains roughly 40% of the observed warming trend, consistent between the neural network and wind-nudged approaches. The linear method consistently estimates a smaller contribution than the machine learning approach, suggesting nonlinear relationships matter for isolating this effect. When natural variability is removed, the mismatch between models and observations in monthly spring warming largely reconciles. The spatial pattern of observed warming also closely resembles that from experiments isolating atmospheric circulation effects, suggesting models simulate the forced warming pattern reasonably well. However, the specific circulation-temperature patterns linked to observed spring warming are rare across climate models, indicating models may struggle to capture the true strength of Arctic natural variability or that the observed patterns are unusual.

1. Introduction

Arctic amplification (AA) is the enhanced surface warming of the Arctic-mean relative to the global-mean. It represents a fundamental response of the Earth system to external forcing, largely independent of the forcing type (Manabe & Stouffer, 1980; Manabe & Wetherald, 1975; Previdi et al., 2021). The magnitude of enhanced Arctic warming varies markedly by season, reflecting the underlying physical mechanisms that drive AA (Bintanja & van der Linden, 2013; Hahn et al., 2021; Lu & Cai, 2009; Pithan & Mauritsen, 2014; Rantanen et al., 2022; Screen

et al., 2012; Stroeve & Notz, 2018). In addition to externally forced changes, internal variability modulates AA (Gale et al., 2026; Gilbert et al., 2025; Sweeney et al., 2023; Ye & Messori, 2021; Zhou et al., 2024). Disentangling the relative contributions of external forcing and internal variability to seasonal AA is critical for robust model evaluation and improving climate projections.

Many studies have investigated the role of internal variability in shaping seasonal Arctic warming and sea ice loss (Deser et al., 2016; Ding et al., 2019; England et al., 2019, 2025; Gilbert et al., 2025; Kay et al., 2011; Roach & Blanchard-Wrigglesworth, 2022; Siew et al., 2024; Swart et al., 2015; Sweeney et al., 2023; Ye & Messori, 2021). Because the background sea ice state, insolation, and atmosphere-ocean coupling vary strongly over the annual cycle, internally-driven circulation anomalies drive Arctic warming through distinct physical mechanisms in each season (Bintanja & Krikken, 2016; Bintanja & van der Linden, 2013). In winter, when the Arctic Ocean is largely ice-covered, circulation-driven sea ice export and thinning in the Barents-Kara Seas expose open water and thin ice to cold, dry air, driving large upward turbulent heat fluxes from ocean to atmosphere and amplifying warming through the lapse-rate feedback (Liu et al., 2022; Pithan & Mauritsen, 2014; Siew et al., 2024). In summer and autumn, teleconnections from remote modes of variability, including those originating in the tropical Pacific and the Pacific North American pattern, generate anomalous anticyclonic circulation and associated large-scale subsidence that adiabatically warms the lower troposphere and increases downward longwave radiation at the surface, accelerating summer sea ice melt whose stored ocean heat is subsequently released to the atmosphere in autumn (Baxter et al., 2019; Ding et al., 2019; England et al., 2019; Liu et al., 2021; Roach & Blanchard-Wrigglesworth, 2022). In spring, internally-driven circulation anomalies advect warm, moist air into the Barents Sea, increasing downward longwave radiation that is critical for initiating Arctic-wide melt onset and the subsequent ice-albedo feedback (England et al., 2019; Mortin et al., 2016; Stroeve & Notz, 2018).

While significant attention has been given to summer, fall, and winter seasons in this regard, spring has received comparatively limited investigation. This gap in literature is notable because climate models generally simulate weaker spring warming than observed, whereas observations consistently reveal a pronounced peak in Arctic-mean air temperature (SAT) trends in April (Gale et al., 2026; Gilbert et al., 2025; Hahn et al., 2021; Rantanen et al., 2022; Screen et al., 2012). Spring has an outsized impact on annual Arctic warming trends since springtime warming initiates the sea ice-albedo feedback and radiative anomalies have the largest impact year-round (Bintanja & Krikken, 2016). The strong sensitivity of warming in spring to circulation anomalies, operating at the transition between cold-season dynamics and the melt season onset, makes it particularly important to understand the impact of internal variability on spring AA. Accurately representing spring SAT trends, including those driven by internal variability, is therefore crucial for the skillful simulation of the Arctic climate system throughout the year.

A commonly used measure of AA is the ratio of Arctic-mean to global-mean SAT trends. An examination of observed monthly AA reveals a clear April peak during spring (Gale et al., 2026; Gilbert et al., 2025; Hahn et al., 2021; Rantanen et al., 2022), a feature absent in model simulations (Figure 1). In particular, the April AA in all observational data sets lies well outside the 5th–95th percentile range of the ensemble member distribution from all models. Consistent with this, over the period 1980 to 2022 and for the region poleward of 70°N, the observed spring (March to May) AA reached 4.2, substantially exceeding the multi-model mean (MMM) value of 2.7, with less than 4% of model simulations producing an AA greater than the observed value (Gale et al., 2026; Hahn et al., 2021; Rantanen et al., 2022; Sweeney et al., 2023).

Extending the annual-mean machine learning framework of Sweeney et al. (2023) to the spring season, Gale et al. (2026) quantified the contribution of internal variability to the observed trends. They demonstrated that removing the internally-driven component largely resolves the difference in spring AA between observations and the MMM. The inferred internal contribution to Arctic-mean spring warming is further corroborated by an independent dynamical adjustment analysis. Their study also identified a pronounced negative sea level pressure (SLP) trend over northwest Siberia (Stroeve & Notz, 2018), accompanied by anomalously positive SAT trend from the Barents to the East Siberian Seas and across northern Siberia. Although the observed internal SAT–SLP relationship is consistent with simulations, it lies at the extreme of the model distribution, indicating that most simulations have difficulty reproducing the magnitude of internal variability.

Wind-nudging experiments successfully capture the observed Arctic spring warming, highlighting the central role of atmospheric circulation in driving the internally-generated component of warming, primarily through wind-driven meridional heat transport and sea-ice modulation (Gilbert et al., 2025; Roach & Blanchard-

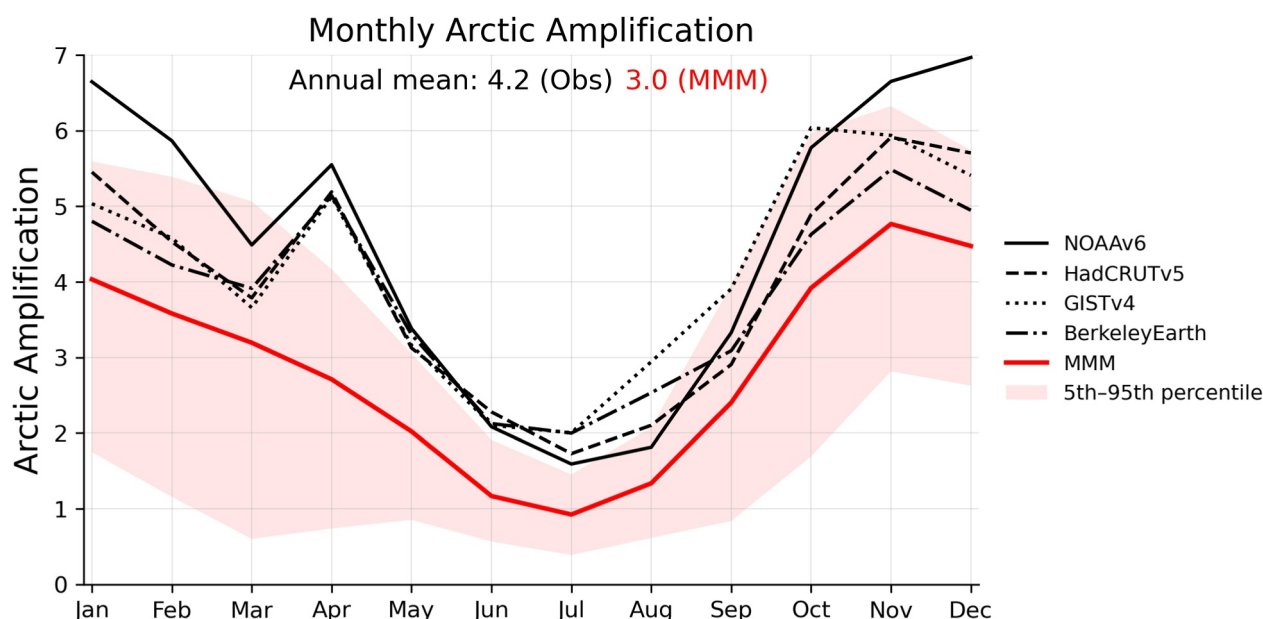


Figure 1. Monthly Arctic amplification (AA) for 1980–2022, shown for four observational surface air temperature (SAT) data sets (black lines) and the multi-model mean (MMM; red line). Light red shading represents the 5th–95th percentile of the trends across all simulated ensemble members. The analysis includes the first eight models listed in Table S1 in Supporting Information S1. Annual-mean values for the observational average and MMM are shown in the upper center.

Wigglesworth, 2022). Based on machine learning-derived internal SAT trends from Sweeney et al. (2023) and wind-driven estimates from Gilbert et al. (2025), about 18% and 15% of the annual Arctic-mean SAT trend for 1980 to 2022 is attributable to internally-driven processes, respectively. In spring, the internally-driven contribution increases substantially, reaching 27% and 24%, according to estimates from Gale et al. (2026) and Gilbert et al. (2025), respectively.

Here, we build on Gale et al. (2026) by examining the role of internal variability in AA for the individual spring months of March, April, and May. Specifically, we address three questions: (1) Do different methodological approaches yield consistent estimates of the internal contribution for each month, as they do for the spring-mean analysis? (2) Does removing the internally-driven component reconcile model simulations with observations on a monthly basis? (3) Does the rare internal SLP–SAT configuration identified in Gale et al. (2026) occur in each spring month? Particular emphasis is placed on April due to its springtime peak in Arctic-mean surface warming and AA. This study advances beyond Gale et al. (2026) in several important respects. First, we present the first comparison of CNN, machine learning using partial least squares (ML-PLS), dynamical adjustment, and wind-nudging estimates for individual spring months. Among these methods, ML-PLS and dynamic adjustment are linear approaches. Second, we reveal that although the dynamical adjustment method closely matches the CNN estimates for the spring-mean trends (Gale et al., 2026), the two methods diverge substantially at the monthly scale, indicating an important role of nonlinear circulation–temperature processes in monthly SAT variability. In contrast, the two linear approaches yield consistent estimates. Third, we identify April as a particularly distinctive month, in which internal variability accounts for approximately 40% of the observed Arctic-mean SAT trend. This result is consistently derived from both the CNN and wind-nudging estimates.

This paper is organized as follows. Section 2 describes the four methods used to estimate internal variability: CNN, dynamical adjustment, ML-PLS, and wind-nudging experiments. Section 3 presents the monthly SAT and SLP trend patterns, decomposes the SAT trend patterns into externally forced and internally-driven components, and examines the associated vertical structures of temperature and zonal wind trends. Section 4 compares estimates of internal trends derived from the CNN, dynamical adjustment, ML-PLS, and wind-nudging experiments. Section 5 examines whether the observed internal circulation–temperature configuration is reproducible within the model ensemble and assesses the extent to which climate models capture the sign and magnitude of the observed internal variability. Finally, Section 6 summarizes the main findings and discusses their physical mechanisms and broader implications.

2. Methods

We apply a suite of complementary approaches to quantify the contribution of internal variability to spring monthly Arctic-mean SAT trends, including statistical and machine learning (ML) techniques, which are comprised of both linear and non-linear approaches. Together, these approaches provide a robust assessment of internal variability and its influence on observed Arctic warming. The observational, reanalysis, and simulation data sets used throughout this study are described in Appendix A.

2.1. Trend and Correlation Analysis

All 43-year linear trends are estimated using ordinary least squares. Correlation is calculated using Pearson correlation coefficient, with statistical significance evaluated using a two-sided Student's *t*-test at the 95% confidence level.

2.2. Convolutional Neural Network (CNN)

To estimate internally-generated and externally forced Arctic- and global-mean SAT trends, we apply CNNs following the framework of Sweeney et al. (2023) and Gale et al. (2026). In this framework, CNNs are applied in a supervised machine learning task. The CNNs are trained using monthly (March, April, and May) spatial maps of 43-year SAT and SLP trends derived from large-ensemble simulations to predict the internally-generated and externally forced SAT trends averaged over the Arctic (poleward of 70°N) or the globe. The combined use of SAT and SLP trend patterns provides sufficient information to diagnose internal climate variability (Gale et al., 2026; Rader et al., 2022; Sweeney et al., 2023). Trend maps are computed over rolling 43-year windows five years apart within a model's extended period (i.e., 1855–1897, 1860–1902, ..., 2005–2047) to increase the size of the training data set. While the trend maps serve as the predictors, the simulated internally-generated and externally forced Arctic- and global-mean SAT trends are the predictands for the CNNs. The externally forced component is calculated as the ensemble mean SAT trend across all ensemble members of a given model, while the internally-generated component is calculated as the deviation of each ensemble member's SAT trend from the ensemble mean. For models with more than 10 ensemble members, a subset of members is randomly selected to avoid over-representing any single model's effective forcing. See Table S1 in Supporting Information S1 for a list of the models, number of ensemble members, and sampled periods used.

All model output and observational fields are bilinearly regridded to a common $2.5^\circ \times 2.5^\circ$ rectilinear grid using xESMF, an open-source Python package, prior to training. Separate CNNs are constructed for the Arctic and the globe. The Arctic CNN is trained on SAT and SLP trend maps poleward of 20°N to capture relevant mid-latitude circulation known to influence Arctic climate through dynamically induced variability (Deser et al., 2016; Smoliak et al., 2015). The global CNN is trained on global maps of SAT and SLP trends. The CNN outputs are the internally-generated and externally forced SAT trends averaged over either the Arctic (poleward of 70°N) or the globe using cosine-latitude weighting. A schematic of the CNN architecture, including the number of trainable parameters, input field dimensions, number of training samples, and the receptive field of the output layer, is provided in Figure S1 in Supporting Information S1. The Arctic CNN takes a $28 \times 144 \times 2$ input array representing SAT and SLP trend maps poleward of 20°N; the global CNN takes a $72 \times 144 \times 2$ input array. Both networks consist of a single Conv2D layer with 16 filters and a 1×3 kernel, followed by MaxPooling2D (pool size 1×1), Dropout ($p = 0.5$), and a Flatten layer. ReLU activation is applied after the convolutional layer. The output is a Dense (2) layer with linear activation.

Model training and evaluation follow a leave-one-model-out cross-validation framework, in which CNNs are trained on multiple models and tested on one withheld model (Gale et al., 2026; Sweeney et al., 2023). The CNN loss function is mean squared error (MSE). Performance is assessed using Pearson correlation coefficients (*r*) and root-mean-square error (RMSE) between predicted and known internal, external, and total (internal plus external) trend components, with particular emphasis on predicting the internal component. After cross-validation, the trained CNNs are applied to observed SAT and SLP trend maps to estimate observed internally-generated and externally forced Arctic- and global-mean SAT trends. The final observed estimates are the average output of the trained cross-validated CNNs. Each spring month is treated independently throughout. See Figures S2–S4 in Supporting Information S1, respectively, for March, April, and May scatter plots of CNN-predicted versus actual values of internal, external, and total trend components. The total uncertainty in the CNN-predicted observational estimate is calculated as the root-sum-square of the standard deviation of the leave-one-model-out CNN estimates

(algorithm uncertainty) and the standard deviation across observational SAT data sets (observational uncertainty). The algorithm uncertainty dominates the total uncertainty.

2.3. Dynamical Adjustment

We apply a dynamical adjustment to estimate the contribution of atmospheric circulation variability to monthly spring Arctic-mean SAT trends. This approach follows the framework introduced in Wallace et al. (2012) and Smoliak et al. (2015), in which large-scale atmospheric circulation is represented by SLP fields and its influence on Arctic-mean SAT is quantified using partial least squares (PLS) regression. The adjustment is applied to monthly Arctic-mean SAT time series, using monthly-mean SLP fields poleward of 40°N to capture large-scale circulation features that modulate Arctic surface temperature variability (Smoliak et al., 2015; Wallace et al., 2012). The dynamically induced monthly SAT estimates have been scaled using the ratio of the observed Arctic-mean SLP trend with the MMM removed to the total observed Arctic-mean SLP trend for 1980 to 2022 to account for any forced contribution in observed SLP. Separate adjustments are applied to March, April, and May. Following a conservative selection criterion, we retain the leading PLS predictors that together account for roughly 40% of the detrended Arctic-mean SAT variance, consistent with previous studies (Gale et al., 2026; Smoliak et al., 2015; Wallace et al., 2012).

2.4. Machine Learning Partial Least Squares (ML-PLS)

To provide a linear ML benchmark, we apply a ML implementation of partial least squares (ML-PLS), following the approach of Po-Chedley et al. (2022). ML-PLS uses separate monthly (March, April, and May) Arctic SAT and SLP trend maps as predictors and the monthly internally-generated and externally forced Arctic-mean SAT trends as predictands, consistent with the CNN framework. The ML-PLS algorithm identifies linear relationships between the predictors and predictands by computing coefficient maps that describe how spatial patterns of SAT and SLP trends map onto internal and forced SAT variability. Apart from the use of a linear algorithm, the training, validation, and application procedures mirror those used for the CNN. PLS regression requires a selection of the number of components (Po-Chedley et al., 2022). To decide how many components are used for each month, a similar leave-one-out cross validation is implemented for different numbers of components, and the value with the smallest RMSE is used. Depending on the month, three to four components are used for ML-PLS. Both dynamical adjustment and ML-PLS are computationally efficient and straightforward to interpret, since the regression coefficients linearly represent the sensitivity of Arctic-mean SAT to the leading modes of SLP variability. However, these methods are linear and additive, providing complementary approaches to the CNN. In particular, ML-PLS provides a transparent linear baseline since it uses the same predictors, training periods, and evaluation framework as the CNN. This allows us to quantify the additional value gained by incorporating nonlinear relationships in the CNN approach.

2.5. Wind-Nudging Experiments

To understand the role of atmospheric circulation, we additionally analyze wind-nudged simulations from CESM2 described in Gilbert et al. (2025). We use the output from the three “PINudge” ensemble members. PINudge consists of CESM2 pre-industrial control simulations where horizontal winds are nudged toward ERA5 reanalysis above 850 hPa from 60°N to 90°N from 1980 to 2023. Nudging is applied only to the free troposphere, allowing surface–atmosphere interactions and thermodynamic feedbacks to evolve freely. The PINudge simulations contain no time-varying external forcings. As a result, these simulations quantify the influence of observed winds on the Arctic system. The PINudge simulations are useful in isolating the wind-driven component of Arctic variability, which can be compared to estimates of the impact of internal variability derived from machine and statistical learning methods.

3. Monthly Spring SAT and SLP Trend Patterns

The observed Arctic trends in SAT and SLP from 1980 to 2022 for March, April, and May are shown in Figure 2. March exhibits the largest circulation anomaly trends, characterized by a negative SLP trend as large as -3.1 hPa/decade over northwest Siberia (Figure 2d), while April shows the strongest temperature anomaly trend as large as 1.7 K/decade, with warming extending from the Barents Sea to the East Siberian Sea and spanning across northern Siberia (Figure 2b). These results point to a previously underemphasized seasonal offset, with circulation trends

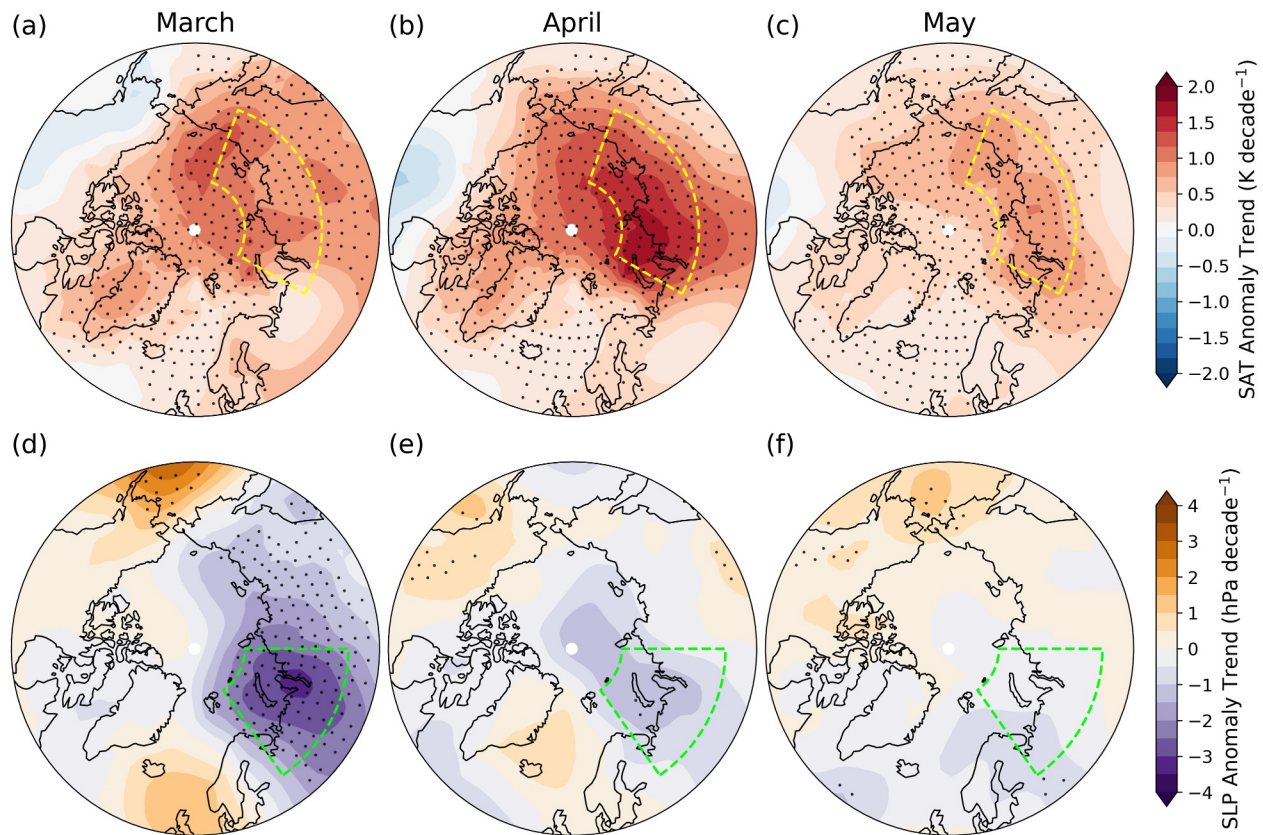


Figure 2. Observed monthly SAT (top row) and SLP (bottom row) trends from 1980 to 2022 for (a, d) March, (b, e) April, and (c, f) May. The dashed yellow box denotes the region of largest SAT anomalies, most prominent in April, while the dashed green box denotes the region of largest SLP anomalies, most prominent in March. Observed SAT and SLP trends are averaged from four observational SAT data sets and three reanalyses, respectively (Appendix A). Stippling indicates regions where trends are significant at the 95% confidence level, accounting for temporal autocorrelation.

peaking in March and surface warming peaking in April. The strong April warming is also reflected in the spring peak AA within the seasonal cycle (Figure 1). May trends in both SAT and SLP are much weaker, reflecting the transition from late spring to early summer conditions. These large month-to-month differences suggest that the influence of atmospheric circulation on Arctic surface temperatures varies monthly, and that averaging over the entire spring season could obscure important variations in circulation-driven warming and cooling.

To assess the impact of month-to-month circulation variability on regional SAT trends, we first turn to the PInudge experiment from Gilbert et al. (2025), which isolates the contribution of circulation anomalies to Arctic SAT trends. The MMM panels (Figures 3a–3c) represent the simulated SAT response to external forcing. Figures 3d–3f shows observations minus MMM. The PInudge panels (Figures 3g–3i) isolate the SAT response to circulation changes, which are primarily driven by internal variability (Gale et al., 2026). A strong resemblance emerges between the difference patterns (Figures 3d–3f) and the PInudge trends (Figures 3g–3i), particularly in April and May, with area-weighted pattern correlations (poleward of 60°N) of 0.80 and 0.76, respectively. In March, both the difference and PInudge maps show cooling over northwest Siberia and warming over eastern Siberia, associated with a cyclonic anomaly centered just south of the Kara Sea (Figure 2d). In April, both also exhibit pronounced warming extending from the Barents Sea to the East Siberian Sea and across northern Siberia. The close agreement between the difference patterns and the circulation-driven PInudge response indicates that the MMM largely captures the externally forced SAT pattern. The high spatial similarity between the observations minus PInudge differences for March, April, and May (Figures S5a–S5c in Supporting Information S1) and the MMM (Figures 3a–3c) further support the point that the forced SAT pattern is well simulated.

We further examine the latitude-height structure of the 1980 to 2022 temperature trends between 60°N and 90°N based on observations, MMM, observations minus MMM, and the PInudge across each spring month (Figure 4). Much of the stratospheric cooling trend seen in observations (Figures 4a–4c) is captured in PInudge

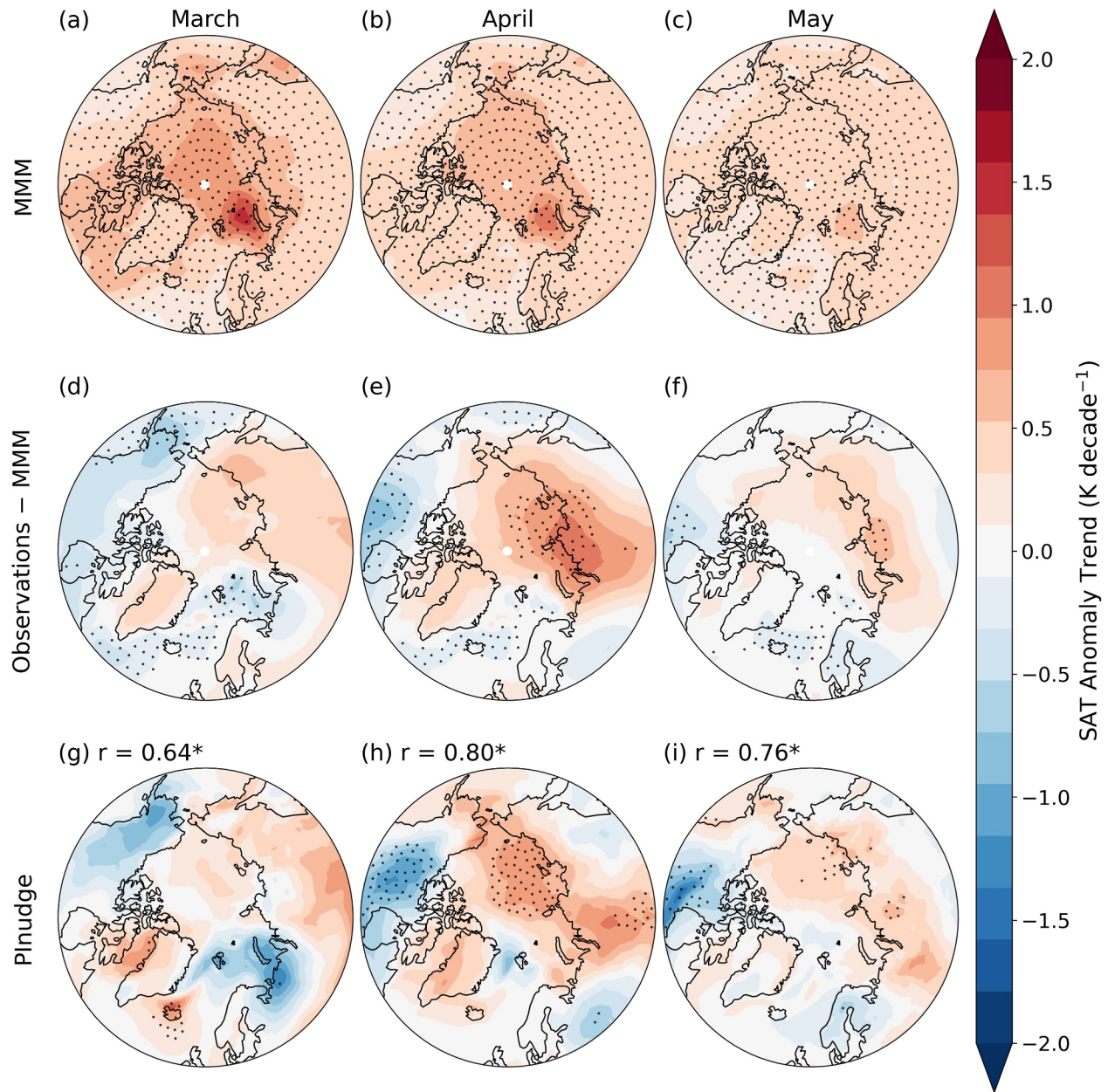


Figure 3. SAT trend maps from 1980 to 2022 for March (left column), April (middle column), and May (right column) from MMM (a–c), observations minus MMM (d–f), and Plinudge experiment (g–i). To account for model climate sensitivity biases, simulated SAT trends are scaled by the ratio of the observed externally forced annual-global-mean SAT trend (0.213 K/decade to the corresponding MMM global-mean SAT trend for 1980–2022 (Gale et al., 2026; Sweeney et al., 2023)). Area-weighted spatial pattern correlations poleward of 60°N between Plinudge and the difference for each month are provided above each corresponding Plinudge panel in the bottom row; an asterisk indicates if it is significant at the 95% confidence level. Stippling indicates regions where trends are significant at the 95% confidence level, accounting for temporal autocorrelation.

(Figures 4j–4l) for each month. For April, the observed warming structure in the troposphere (Figure 4b) is also reflected in the difference with the MMM (Figure 4h) and Plinudge (Figure 4k). Because Plinudge is constrained by the observed atmospheric circulation, this suggests that the depth of the observed tropospheric warming in April is partly related to atmospheric dynamics. The MMM (Figures 4d–4f) demonstrates a surface-based, more uniform vertical warming pattern, also seen in the difference between observations and Plinudge (Figure S6 in Supporting Information S1). This consistency between the MMM and the Obs–Plinudge residual supports the interpretation that, once the circulation-driven component of the observed trend is removed, the remaining pattern resembles the externally forced warming captured by the model simulations.

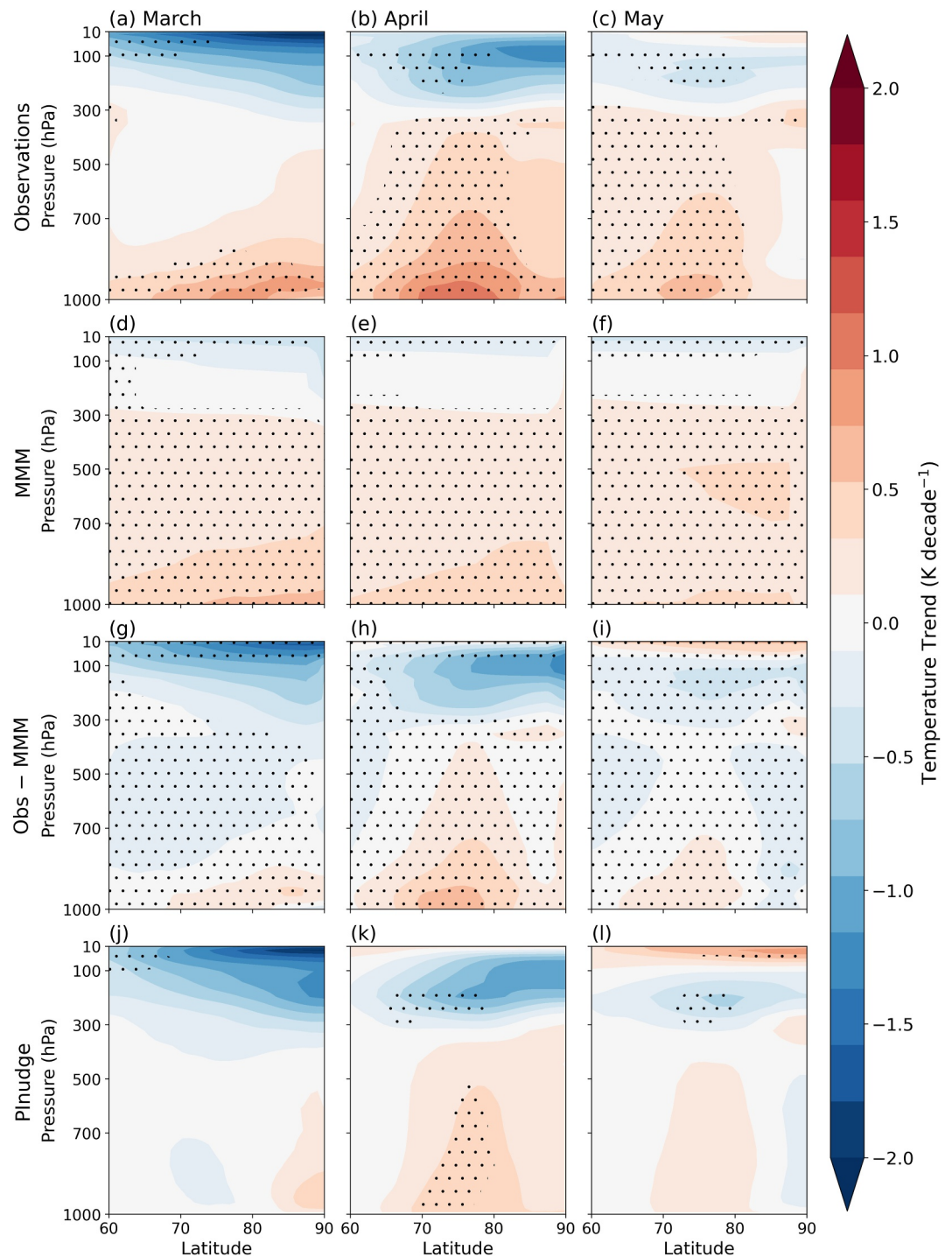


Figure 4. Temperature trends for 1980–2022 as a function of pressure and latitude from 60 to 90°N for March (left column), April (middle column), and May (right column) from observations (ERA5) (a–c), MMM (d–f), observations minus MMM (g–i), and PINudge (j–l). Stippling indicates regions where trends are significant at the 95% confidence level, accounting for temporal autocorrelation.

Figure 5 shows the zonal-mean zonal wind trend for observations and the MMM. The observed zonal wind trend pattern (Figures 5a–5c), including its enhanced upper-tropospheric magnitude and sign, is consistent with the temperature trend pattern in PINudge (Figures 4j–4l), as expected through the thermal wind balance. By contrast, the MMM shows little change in zonal wind (Figures 5d–5f). This difference suggests that the observed

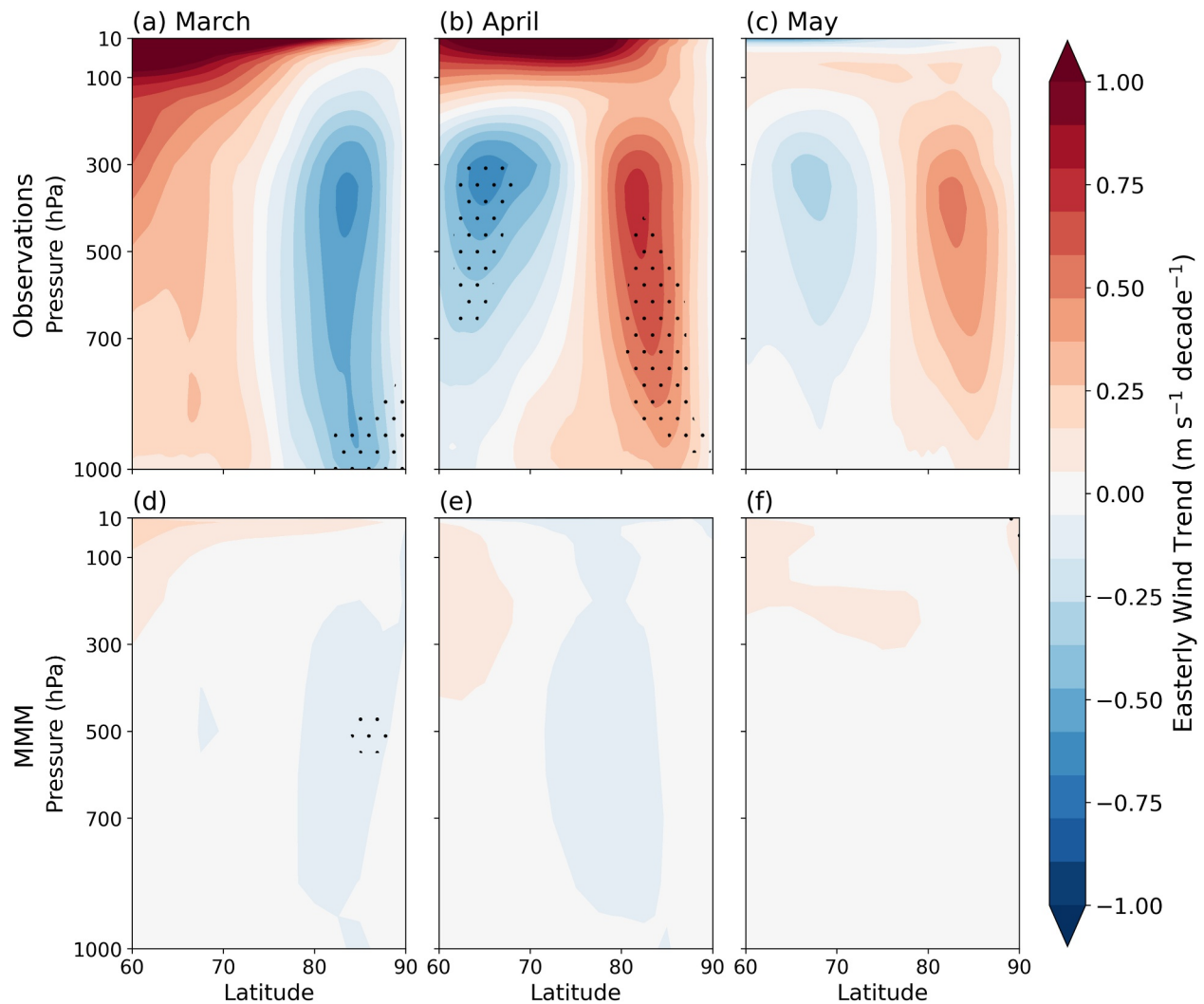


Figure 5. Zonal-mean zonal wind trend for 1980–2022 as a function of pressure and latitude from 60°N to 90°N for March (left column), April (middle column), and May (right column) from observations (ERA5) (a–c) and MMM (d–f). Stippling indicates regions where trends are significant at the 95% confidence level, accounting for temporal autocorrelation.

enhancement of upper-tropospheric wind trend is not part of the externally forced response captured by the models, and instead reflects internally generated atmospheric variability. Together, these diagnostics provide dynamical context for the surface warming patterns shown in Figures 2 and 3 and help motivate the monthly-scale analysis by highlighting physical distinctions between months.

4. Internal Variability and Monthly Arctic Amplification

Table 1 summarizes the observed SAT trends (K/decade) and AA values from 1980 to 2022 for March, April, and May, alongside MMM estimates and observed AA after removing internal components (i.e., observed externally forced AA). Internal contributions are derived using two methods: CNN and ML-PLS. The table also includes monthly Arctic-mean internal estimates from the dynamical adjustment and the Gilbert et al. (2025) Plunge experiment. Observed AA is largest in April (5.3), with lower values in March (4.0) and May (3.2), reflecting both the springtime monthly evolution of Arctic warming and the relative influence of internal variability during each spring month. The MMM provides a modeled estimate of externally forced AA of 3.2 [2.2, 4.4] in March, 2.7 [1.9, 3.6] in April, and 2.0 [1.7, 2.5] in May, which are consistently lower than the corresponding observed total AA, particularly in April. The intermodel spread in the externally forced SAT trend and AA reflects structural

Table 1

Arctic- and Global-Mean Monthly SAT Trends (K/decade) and AA From 1980 to 2022 for March, April, and May

		March	April	May
Observed	Arctic	0.797	1.004	0.558
	Global	0.201	0.191	0.172
	AA	4.0	5.3	3.2
MMM	Arctic MMM	0.671 ± 0.553	0.551 ± 0.390	0.397 ± 0.194
	Global MMM	0.206 ± 0.096	0.202 ± 0.088	0.197 ± 0.084
	Forced AA	3.2 [2.2, 4.4]	2.7 [1.9, 3.6]	2.0 [1.7, 2.5]
CNN	Arctic Internal	0.156 ± 0.416	0.397 ± 0.312	0.171 ± 0.172
	Global Internal	0.002 ± 0.048	-0.001 ± 0.044	-0.007 ± 0.038
	Forced AA	3.2 [1.8, 4.8]	3.2 [2.0, 4.4]	2.2 [1.2, 3.2]
ML-PLS	Arctic Internal	0.125 ± 0.196	0.253 ± 0.154	0.085 ± 0.070
	Global Internal	-0.003 ± 0.010	-0.014 ± 0.006	-0.019 ± 0.002
	Forced AA	3.3 [1.8, 4.8]	3.7 [2.3, 5.1]	2.5 [1.5, 3.5]
Dynamical Adjustment (Scaled)	Arctic Internal	0.116 ± 0.252	0.250 ± 0.308	0.085 ± 0.102
PIinudge ^a	Arctic Internal	0.004 ± 0.150	0.391 ± 0.016	0.168 ± 0.137

Note. AA is shown from observations, while the forced AA is shown from MMM and from observations with internal trends removed by the CNN and ML-PLS methods. Arctic- and global-mean internal contributions to observed SAT trends are reported separately, with additional Arctic-mean internal estimates from dynamical adjustment and PIinudge experiment. Uncertainties correspond to two standard deviations, except for AA which is reported as [5th, 95th] percentile range.

^aThe Arctic-mean SAT trends from ERA5 for 1980–2022 are 0.633 K/decade in March, 0.888 K/decade in April, and 0.411 K/decade in May. PIinudge used solely ERA5 data, so for a direct comparison with other methods used in this study, the PIinudge internal trends are scaled by the ratio of observed to ERA5 Arctic-mean SAT trends for each respective month.

differences across models, including differences in physical parameterizations, and is reported as percentile ranges in Table 1.

Estimates from all approaches (except PIinudge in March) indicate that internal variability contributes substantially to Arctic-mean spring monthly warming over 1980 to 2022. The CNN-derived internal trends are 0.156 K/decade in March, 0.397 K/decade in April, and 0.171 K/decade in May, accounting for about 20%, 40%, and 31% of the observed Arctic-mean SAT trends, respectively. The CNN, ML-PLS, and dynamical adjustment estimates are broadly consistent, with nearly identical estimates from ML-PLS and dynamical adjustment (Table 1), lending confidence to the methodology and data sets used to train and validate the ML framework. The CNN yields larger internal contributions than ML-PLS, especially in April, suggesting an important role of nonlinear relationships between monthly circulation anomalies and temperature responses. However, the close agreement between CNN and dynamical adjustment estimates for the spring-mean (Gale et al., 2026) indicates that such nonlinear effects are likely of secondary importance in seasonal-mean analyses.

The circulation-induced SAT trends based on the PIinudge simulations are 0.391 K/decade in April and 0.168 K/decade in May, in close agreements with the CNN-derived internal trends of 0.397 and 0.171 K/decade, respectively (Table 1). In contrast, the PIinudge SAT trend is near-zero in March. The difference may be linked to March-PIinudge's regional biases in the Barents Sea, where nudged simulations tend to underestimate winter sea ice loss, potentially leading to excessive March cooling (Gilbert et al., 2025; Roach & Blanchard-Wrigglesworth, 2022). Uncertainties associated with the nudging framework and the imposed ERA5 winds may also contribute. In March, nonlinear effects appear relatively weak, with consistent and comparable Arctic-mean internal warming estimates of 0.116 (15%), 0.125 (16%), and 0.156 (20%) K/decade from dynamical adjustments, ML-PLS, and CNN, respectively. Overall, Table 1 demonstrates pronounced month-to-month differences in the internal contributions to spring AA.

Internal variability is most influential in April when circulation-driven warming is strongest, while March and May show more modest internally driven contributions. However, circulation trends are weaker in April than in March, indicating that the internally-driven temperature response is also dependent on the seasonal timing and the pattern of the circulation anomalies. After removing the internal components from the observed trends, the CNN-

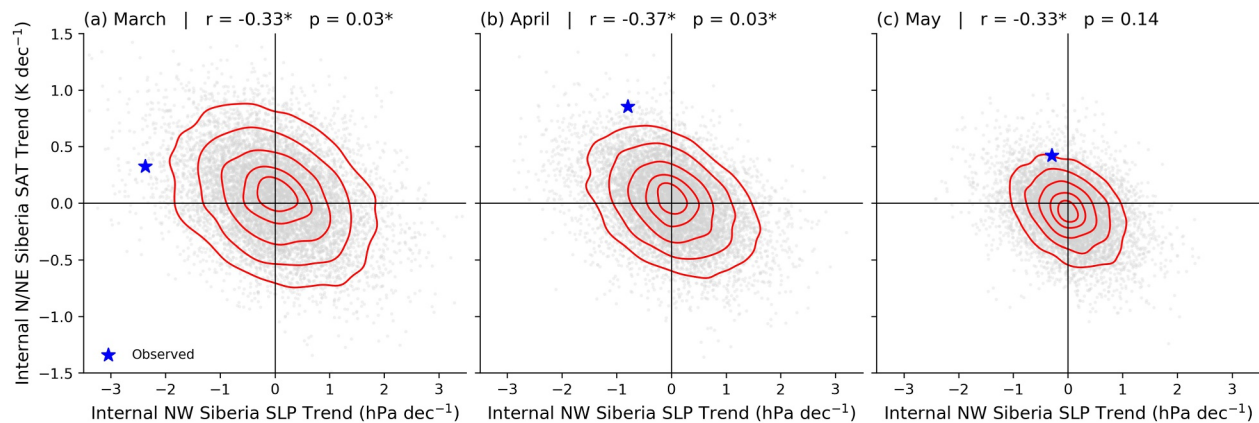


Figure 6. Scatter plots of regional-mean internal SLP trends (x-axis) versus internal SAT trends (y-axis). The gray dots represent all simulated 43-year trends for (a) March, (b) April, and (c) May, across all model ensemble members and periods (Table S1 in Supporting Information S1) with 5 years interval, totaling 9,866 points. Contours depict the normalized kernel density estimate of the simulated trend pairs, corresponding to 10%, 25%, 50%, 75%, and 90% of the peak density. The blue star denotes the observed internally-driven trend pair estimated from the difference between observations and the MMM for the specified regions (Figure 2). The correlation of the simulated points and the Mahalanobis outlier test p -value are shown above each panel; an asterisk is included if it is significant at the 95% confidence level.

based observed forced AA is 3.2 [1.8, 4.8] in March, 3.2 [2.0, 4.4] in April, and 2.2 [1.2, 3.2] in May. These values agree with corresponding AAs from MMM within the uncertainty range, although the MMM may still somewhat underestimate externally forced AA in April.

5. Likelihood of Observed Internal Atmospheric Circulation-Temperature Pairing in Simulations

We further investigate regions of largest trend in observed circulation (northwest Siberia; 60°N–80°N, 35°E–90°E) and SAT (north/northeast Siberia; 65°N–80°N, 60°E–160°E) as shown by green and yellow boxes in Figure 2, respectively. Figure 6 shows the relationship between simulated internal SLP and SAT trends averaged over these regions. The gray dots represent all simulated 43-year trends for March, April, and May across all model ensemble members and periods (Table S1 in Supporting Information S1), totaling 9,866 points. Red contours show the relative density of simulated internally driven trend pairings. The blue star marks the observed internally driven trend pair, approximately estimated as the difference between observations and the MMM. For reference, the regional-mean observed (MMM) SLP trends are -2.55 (-0.18) hPa/decade, -1.00 (-0.20) hPa/decade, and -0.33 (-0.04) hPa/decade for March, April, and May, respectively. The regional-mean observed (MMM) SAT trends are 0.934 (0.614) K/decade, 1.384 (0.537) K/decade, and 0.784 (0.368) K/decade for March, April, and May, respectively. Note that the forced circulation trends are minimal (see Figure S7 in Supporting Information S1).

Here we assume that the observed-minus-MMM difference primarily reflects internal variability. Confidence in the MMM is supported by its realistic representation of monthly externally forced AA and the associated warming pattern (Table 1; Figure 3). Consistent with this interpretation, the differences between observed and MMM Arctic-mean trends (0.126 , 0.453 , and 0.161 K/decade for March, April, and May, respectively; calculated from Table 1) closely align with the internally driven Arctic-mean trends estimated by the CNN (0.156 , 0.397 , and 0.171 K/decade; Table 1). Taken together, the consistent externally forced MMM AA, the high spatial correlations between Obs–MMM and Pinudge, and the agreement with CNN-based internal estimates provide complementary and mutually reinforcing evidence supporting the MMM as a reliable estimate of the forced response.

Figure 6 demonstrates how frequently particular combinations of internally driven circulation and temperature trends occur in the models. In March, the observed point lies at -3σ along SLP ($\sigma = 0.9$ hPa/dec) and 1σ along SAT ($\sigma = 0.4$ K/dec), corresponding to 0.7% and 23% of simulated values larger than observational magnitudes, respectively, and 0.4% of simulations simultaneously larger than both observed SLP and SAT trend magnitudes. This indicates that the strong cyclonic circulation over northwest Siberia combined with the associated magnitude of SAT anomalies is highly unusual in models but still within the simulated range of internal variability. In April, the observed point is -1σ along SLP ($\sigma = 0.8$ hPa/dec) and 3σ along SAT ($\sigma = 0.3$ K/dec), with 14% and 0.5% of

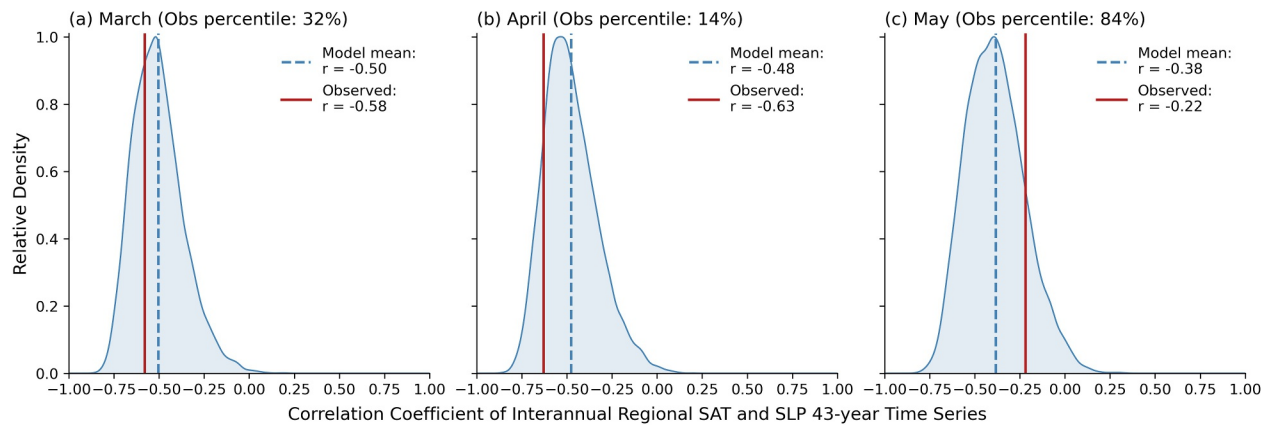


Figure 7. Probability density functions of correlation coefficients (r) of detrended SLP and SAT 43-year time series, averaged over the northwest Siberia and north/northeast Siberia domains (Figure 2), respectively, for (a) March, (b) April, and (c) May, from model simulation across all model ensemble members and all 43-year periods (sampled with starting years separated 5 years apart), totaling 9,866 pairs. Dashed vertical blue lines show the MMM correlation coefficient for each month. Vertical red lines show the observed detrended SLP–SAT correlation for the corresponding month for 1980–2022. Percentiles indicate the fraction of simulated timeseries correlation coefficients that are less than or equal to the observed value.

simulations exceeding these values individually and only 0.3% of simulations exceeding both, showing that the strong April circulation-driven warming is rare while still reproducible, particularly along the circulation axis. For May, the observed trends are -1σ in SLP ($\sigma = 0.6$ hPa/dec) and 2σ in SAT ($\sigma = 0.3$ K/dec), with 26% and 3% individually exceeding and 1% jointly exceeding, indicating a moderate rarity of the May configuration. Taken together, these results indicate that the observed spring internal circulation–temperature configuration is rare in the models.

To further assess the joint rarity of the observed SLP–SAT pairing in a multivariate test, we compute the squared Mahalanobis distance (D^2) between the observed trend pair and the mean of the simulated distribution. D^2 generalizes the squared z-score to multiple, correlated variables, measuring the squared distance between a point and the distribution's mean while accounting for the covariance between variables. The D^2 values are 6.8, 7.0, and 3.9 for March, April, and May, respectively. The resulting p-values are 0.03*, 0.03*, and 0.14 for March, April, and May, respectively, where an asterisk indicates significance at the 95% confidence level. This confirms that the observed configurations for March and April are statistically unlikely to be drawn from the distribution from the model simulations. However, the observed configuration remains a feature of simulated internal variability, representing an extreme but physically plausible realization that models can capture, albeit infrequently. We obtained similar results using regionally averaged internal trends from PINudge (see Figure S8 in Supporting Information S1). This further supports the finding that observed internally driven springtime monthly SLP and SAT trends rarely occur in simulations.

To assess whether the trend-based SLP–SAT relationship shown in Figure 6 reflects a robust physical coupling, we examine the interannual connection between detrended regional-mean SLP and SAT anomalies over the same domains in both the observations and the model simulations. The probability density functions of the correlation coefficients are shown in Figure 7, based on detrended 43-year time series from all model ensemble members and periods (Table S1 in Supporting Information S1) sampled at 5-year intervals. The resulting mean correlation coefficients are -0.50 , -0.48 , and -0.38 for March, April and May, respectively, comparable to those obtained from the 43-year trends (Figure 6). Furthermore, the probabilities of obtaining positive correlation coefficients are only 0.2%, 0.4% and 1.5% for March, April, and May, respectively, indicating a consistently negative SLP–SAT relationship across the model simulations.

The observed correlation coefficients on the interannual timescale over 1980 to 2022 are -0.58 , -0.63 , and -0.22 for March, April, and May, respectively, corresponding to the 32nd, 14th, and 84th percentile of the model-derived distributions. Thus, the observed correlations lie well within the range simulated by the models. These results indicate that the regional SLP–SAT relationship is robust on both interannual and decadal timescales in observations and model simulations. They further strengthen our confidence that the observed rare circulation–temperature configurations identified in Figure 6, despite their extreme magnitude, are physically

grounded in year-to-year variability. We also examine the results corresponding to Figure 7 using the 43-year time series of model interannual variability (i.e., the difference between each ensemble member and the corresponding ensemble mean) (not shown). The resulting PDF and its associated characteristics are nearly identical to those shown in Figure 7.

6. Conclusions

This study demonstrates that the contribution of internal variability to recent spring Arctic-mean SAT trends is strongly dependent on the analyzed month. Most notably, internal variability explains approximately 40% of the observed April Arctic-mean SAT trend during 1980 to 2022, a result that is consistent across ML-based CNN attribution and wind-nudged simulation methods. The agreement between these fundamentally different methods, supported by analysis of vertical temperature trend structure, zonal wind trends, and interannual variability correlation, provides strong evidence that a substantial contribution of April Arctic warming arises from internally-generated atmospheric variability. This finding highlights the importance of resolving individual spring months rather than relying on seasonal-means alone: the April peak in internally-driven warming is obscured when March, April, and May are averaged, and the processes responsible for April's distinctiveness, including its relatively weak SLP yet anomalously large SAT response, would remain unidentified in a spring-mean analysis.

For each spring month, the linear methods (ML-PLS and dynamical adjustment) yield consistent estimates of internal variability, but their contributions are systematically smaller than those derived from the CNN. The distinct monthly behavior revealed here, whereby the linear dynamical adjustment method agrees with the CNN for spring-mean trend (Gale et al., 2026) but diverges substantially at the monthly scale, suggests that nonlinear circulation–temperature interactions may become important at subseasonal timescales. Additionally, nonlinear approaches may more effectively isolate the internally-driven component. This interpretation is supported by out-of-sample tests in which the CNN, when applied to extremely high spring-mean AA cases not included in training, successfully recovers the simulated trends (not shown).

Removing internal variability largely reconciles the monthly AA discrepancy between models and observations in spring. Furthermore, the pattern difference between observations and the MMM is strongly correlated with the wind-nudged simulations over 60°N polewards across March, April and May. The highest correlation occurs in April ($r = 0.80$, $p = 0.0003$). This indicates that much of the observed departure in spring AA from the MMM can be explained by circulation anomalies present in the real atmosphere but largely absent from the forced response. Additionally, since the residual pattern is well captured when observed winds are imposed, the underlying forced response represented by the MMM is physically consistent. These results support the interpretation that models accurately simulate the externally forced component of springtime monthly AA and warming pattern. This has important implications for near-term projections: differences between observed and simulated spring Arctic warming appear to arise from internal variability rather than errors in the forced response, suggesting that model–observation differences in spring AA will narrow with the reversal of the observed manifestation of internal variability, rather than persisting as a systematic model bias.

While the models reproduce the externally forced signal, the observationally based internal Siberian SAT and SLP trends lie near the edge of the simulated SAT–SLP relationship distribution, particularly in March and April. This suggests that the observed spring circulation–temperature configuration is rare in models but remains within the range of simulated internal variability, consistent with an extreme but plausible real-world realization. The atmospheric circulation anomalies identified here, particularly the persistent low-pressure anomaly over north-west Siberia, represent a physically motivated target for process-based seasonal prediction of spring Arctic warming, since the rarity of the observed configuration in model simulations indicates that accurately representing the statistics of internal variability extremes remains a key challenge for decadal prediction. The extent to which this reflects an unusually rare real-world event versus a potential underrepresentation of such extremes in models remains unclear.

Although this study quantifies the magnitude of the contribution of internal variability to observed SAT trends, it does not directly identify the physical mechanisms that generate or amplify these internally-driven trends. The results are consistent with a framework in which atmospheric circulation anomalies interact with cloud, moisture, and/or surface albedo feedbacks to modulate the timing and strength of Arctic warming, but the specific pathways linking circulation anomalies to radiative and moisture processes remain unclear. It is particularly noteworthy how large the internally-generated warming in April is, given the relatively weak circulation trends during that

month. Future work should explicitly connect circulation-driven variability to changes in cloud properties, atmospheric moisture transport, and the surface energy budget (i.e., snow cover loss), and assess whether model biases in these processes contribute to the apparent underestimation of internal variability. More broadly, these results motivate extending the monthly-scale attribution framework to other seasons and pairing it with process-level diagnostics to directly connect circulation anomalies to the surface energy budget responses that drive AA. A process-based understanding of these interactions is essential for improving prediction capability in the Arctic system from subseasonal to decadal timescales.

Across all spring months, internal variability has accelerated Arctic warming in the recent observational record. However, because modes of internal variability fluctuate between positive and negative phase, a future reversal could lead to a slowdown of spring warming rates. Given the importance of spring radiative anomalies on annual-mean Arctic warming (Bintanja & Krikken, 2016), such a reversal could meaningfully alter year-round warming trajectories, complicating near-term Arctic projections. More broadly, these results underscore that accurately characterizing internal variability is essential not only for interpreting past Arctic warming but also for projecting future Arctic surface warming.

Appendix A: Data

In this study, observed 2-m SAT is from the Berkeley Earth Land/Ocean Temperature Record (BerkeleyEarth) (Rohde & Hausfather, 2020), GISS Surface Temperature Analysis version 4 (GISTv4) (Lenssen et al., 2019), Met Office Hadley Centre/Climatic Research Unit's version 5 (HadCRUTv5) (Morice et al., 2021), and NOAA Merged Land Ocean Analysis version 6 (NOAAv6) (Huang et al., 2024). Observed SLP is from the fifth-generation European Centre for Medium-Range Weather Forecasts Reanalysis (ERA5) (Hersbach et al., 2020), Japan Meteorological Agency 55-year Reanalysis (JRA-55) (Kobayashi et al., 2015), and Modern-Era Retrospective Analysis for Research and Applications version 2 (MERRA-2) (Gelaro et al., 2017).

The model data is based on output from 11 models from the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016), supplemented with E3SMv2 (Fasullo et al., 2024; Golaz et al., 2022) and the smoothed biomass burning aerosol emission version of CESM2 (Fasullo et al., 2022). These models were chosen as they have 10 or more available ensemble members with monthly SAT and SLP fields for the historical period and compatible future scenario, suitable for training machine learning on large ensembles. Ensemble sizes and period for the models used are listed in Table S1 in Supporting Information S1. For eight models, an extended period is created by splicing historical and future scenario data (SSP3-7.0 or SSP5-8.5; SSP5-8.5 is used if both are available). The remaining four models only have sufficient ensemble members until 2014, and thus only periods from 1855 to 2012 are used to train our machine learning (ML) algorithm for these models. For the model analyses, the ensemble means (mean across ensemble members within a model) are interpreted as the simulated response to external forcing, and deviations of individual ensemble members from the ensemble mean are interpreted as internally generated variability (Kay et al., 2015; Sweeney et al., 2023). Like Sweeney et al. (2023), ML training and evaluation are performed using large-ensemble simulations, defined here as models with 10 or more ensemble members. Models with fewer than 10 ensemble members are retained (grouped in OthersAllEM) in order to illustrate inter-model spread but are not used for training. We additionally use wind-nudged simulations from CESM2 described in Gilbert et al. (2025); see Section 2.5 for details.

All observed, reanalysis, and simulated fields are bilinearly regridded to $2.5^\circ \times 2.5^\circ$ resolution. Monthly anomalies are defined relative to each data set's 1980 to 2022 climatology.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The processed data used in this study is archived at Zenodo (Gale, 2026a) at <https://doi.org/10.5281/zenodo.18842088> under a CC BY 4.0 license. Version 2.0 of the code used to preprocess the data, set up, train the CNN, ML-PLS, dynamical adjustment, and reproduce the figures is archived at Zenodo (Gale, 2026b) at <https://doi.org/10.5281/zenodo.20040165> under the MIT License and is also available for development at <https://github.com/sky>

largale/arctic-monthly-internal. Additional data exceeding Zenodo's size limits, including monthly trend files, spliced data sets, and CNN training data are available at the National Center for Atmospheric Research GLADE Globus Share (Gale, 2026c) at https://app.globus.org/file-manager?origin_id=6d205bd9-0908-4092-bda6-ad27854fb59d&origin_path=%2F, no registration or account required. Pinudge simulation output, produced by Gilbert et al. (2025), is hosted in the same Globus Share with permission; see Gilbert et al. (2025) for the original archive and citation.

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